# Tutorial - 3.

Q. Remove left recursion.

① $E \rightarrow E+T \mid D$

$\rightarrow E \rightarrow DE'$

$E' \rightarrow +TE' \mid \epsilon$

② $E \rightarrow E+T \mid E-T \mid T$

$\rightarrow E \rightarrow TE'$

$E' \rightarrow +TE' \mid -TE' \mid \epsilon$

③ $T \rightarrow T*F \mid T/F \mid F$

$\rightarrow T \rightarrow FT'$

$T' \rightarrow *FT' \mid /FT' \mid \epsilon$

④ $F \rightarrow (E) \mid id$

$\rightarrow$ It has no left recursion.

⑤ $A \rightarrow Aa \mid Ab \mid c$

$\rightarrow A \rightarrow cA'$

$A' \rightarrow aA' \mid bA' \mid \epsilon$

⑥ $B \rightarrow Bx \mid By \mid z$

$\rightarrow B \rightarrow zB'$

$B' \rightarrow xB' \mid yB' \mid \epsilon$

⑦ $E \rightarrow E+T \mid E \times T$

$\rightarrow E \rightarrow TE'$

$E' \rightarrow +TE' \mid \times TE' \mid \epsilon$

⑧ $A \rightarrow ABd \mid Aa \mid a$

$\rightarrow A \rightarrow aA'$

$A' \rightarrow BdA' \mid aA' \mid \epsilon$

⑨ $B \rightarrow Be \mid d$

$\rightarrow B \rightarrow dB'$

$B' \rightarrow eB' \mid \epsilon$

Q. Remove left factoring.

① $E \rightarrow T+E \mid T$

$T \rightarrow int \mid int * T \mid (E)$

$\rightarrow E \rightarrow TE'$

$E' \rightarrow +E \mid \epsilon$

$T \rightarrow int\ T' \mid (E)$

$T' \rightarrow *T \mid \epsilon$

② $S \rightarrow aA \mid aB$

$\rightarrow S \rightarrow aS'$

$S' \rightarrow A \mid B$

③ $A \rightarrow abc \mid abd$

$\rightarrow A \rightarrow abA'$

$A' \rightarrow c \mid d$

④ $S \rightarrow$ if E then S else S | if E then S

$\rightarrow S \rightarrow$ if E then S S'

$S' \rightarrow$ else S | $\epsilon$.

5) $B \rightarrow xy \mid xz$

$\rightarrow$ $B \rightarrow xB'$

$B' \rightarrow y \mid z$

6) $A \rightarrow pqr \mid pqs \mid pt$

$\rightarrow$ $A \rightarrow pA'$

$A' \rightarrow qr \mid qs \mid t$

$A' \rightarrow tA'' \mid t$

$A'' \rightarrow r \mid s$

7) $S \rightarrow abc \mid abd \mid ae$

$\rightarrow$ ~~S'~~ $S \rightarrow aS'$

$S' \rightarrow bc \mid bd \mid e$

$S' \rightarrow bS'' \mid e$

$S'' \rightarrow c \mid d$